

THE SILENCE EXPERIMENT

A Multi-State Protocol for Detecting Consciousness-Dependent Decoherence in Shielded Environments

THEORETICAL PHYSICS • CONSCIOUSNESS STUDIES • QUANTUM FOUNDATIONS

By

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May 2026 • Version 10

Paper 2 of 2 • Concept Paper — Not Yet Peer Reviewed

TABLE OF CONTENTS

I. Abstract	
II. Introduction: The Measurement Problem and the Observer	
III. Theoretical Context	
IV. The Mechanism Problem	
V. Experimental Design	
5.1 Multi-State Consciousness Conditions	
5.2 The Shielded Chamber and Five-Layer Isolation	
5.3 Quantum Sensor Options	
5.4 Instrument Array and Environmental Monitoring	
5.5 Session Protocol (Revised)	
5.6 Participants	
5.7 Blinding and Replication	
VI. Primary Experiment: Quantum Decoherence Test	
VII. Secondary Experiment: QRNG Test	
VIII. Data Analysis and Statistical Framework	
IX. Power Analysis	
X. Stopping Rules	
XI. Budget, Phasing, and Timeline	
XII. Ethical Considerations	
XIII. Limitations	
XIV. Future Directions	
XV. Conclusion	
Appendix A — Personal Exploratory Activity (Not Part of the Main Protocol)	
References	

SECTION I

Abstract

This paper proposes a controlled experimental protocol for testing the von Neumann-Wigner (VNW) interpretation of quantum mechanics — the hypothesis that consciousness plays a role in quantum state reduction — against the prevailing environmental decoherence framework (Zurek, Zeh, Joos). The protocol measures whether the cognitive state of a nearby conscious observer measurably affects the decoherence rate of a prepared quantum system under controlled conditions.

The experimental platform places an experienced meditator in EEG-verified states ranging from active cognition to objectless awareness inside a shielded environment. A quantum coherence sensor measures decoherence rates (T_2) across seven controlled conditions spanning the full cognitive-excitation axis. The primary pre-registered test is the ordered prediction $T_{\text{coherence}}(\text{cessation}) > T_{\text{coherence}}(\text{baseline}) > T_{\text{coherence}}(\text{active cognition})$, tested with a pre-registered Jonckheere-Terpstra ordered test. The simple cessation-vs-empty-chamber comparison is a secondary test. All other cross-condition comparisons are treated as exploratory. The multi-state design allows data to determine which consciousness state, if any, produces the largest effect.

A secondary experiment uses quantum random number generators with improved shielding, blinding, and instrumentation.

Phase 0 is a foundational test of the von Neumann-Wigner interpretation under controlled observer states — a question that has been part of the quantum literature for nearly a century without ever being tested this way. The chamber's specifications were dictated by what is required to make any consciousness-dependent T_2 signal interpretable: drive every known environmental coupling channel below the noise floor and monitor each one synchronously. A sensor-and-shielding characterisation of the NV-center platform under that isolation regime — with quantified upper bounds on residual electromagnetic, thermal, vibrational, and magnetic coupling to T_2 — is produced as an independently publishable co-product of the same hardware run, in a metrology venue. The chamber exists for the consciousness test; the metrology paper is what falls out of doing the consciousness test correctly.

The protocol is phased: a Phase 0 study (~\$420,000–\$470,000) using nitrogen-vacancy centers in diamond, and a Phase 1 full protocol (~\$1M–\$3M) using superconducting qubits in an underground laboratory, conditional on Phase 0 results and adoption by an institutional host. (A small commercial-QRNG exploratory activity is described separately in Appendix A as personal exploratory activity, not part of the main protocol.) Analysis is pre-registered, randomised, outcome-blinded where feasible, and fully blinded by an independent statistician who does not see condition labels until the analysis is locked. Phase 0 can detect medium-to-large effects ($d > 0.3$); it is not designed or powered to test small-effect claims from prior consciousness-RNG literature.

This protocol is proposed as a contribution to an ongoing line of experimental inquiry into the measurement problem in quantum mechanics, not as a definitive resolution. Its value lies in

the specificity of its predictions and the rigor of its controls. Null results establish pre-registered upper bounds; positive results require independent replication.

SECTION II

Introduction: The Measurement Problem and the Observer

Quantum mechanics contains an unresolved foundational question known as the measurement problem. A quantum system exists in superposition — multiple states simultaneously — until it undergoes a transition to a single definite outcome. What causes this transition has been debated since the formalization of quantum theory in the late 1920s and remains unresolved today.

Every major interpretation of quantum mechanics is, at its core, a different answer to this question. The Copenhagen interpretation treats measurement as a primitive concept without further analysis. The Many-Worlds interpretation (Everett, 1957) denies that collapse occurs at all. Bohmian mechanics introduces hidden variables. QBism treats quantum states as subjective degrees of belief. And the von Neumann–Wigner (VNW) interpretation, first articulated by von Neumann in 1932 and extended by Wigner in 1961, proposes that consciousness itself is the agent of state reduction.

The VNW interpretation is not mainstream. It is, however, not empirically refuted. Its marginal position in contemporary physics reflects philosophical discomfort, the success of environmental decoherence theory, and the absence of controlled experiments designed to distinguish between interpretations using the observer's cognitive state as an experimental variable.

2.1 Environmental Decoherence: The Mainstream Position

The environmental decoherence program (Zurek, 2003; Zeh, 1970; Joos et al., 2003) provides a mechanism by which quantum superpositions become effectively classical through interaction with the environment. However, decoherence does not solve the measurement problem in its entirety. As Zurek has acknowledged, decoherence explains the suppression of interference but does not explain why a specific outcome is realized from among the surviving diagonal elements. This residual gap — the problem of definite outcomes — is where the VNW interpretation operates.

2.2 The Boundary Effect

The theoretical framework (Paper 1: The Silence Paradigm) proposes a boundary effect: decoherence is the quantum–classical boundary, and if consciousness participates in state reduction, the effect should manifest at this boundary as a shift in coherence times. This experiment measures not consciousness itself but the shift in the decoherence boundary associated with the observer's state.

2.3 The Experimental Gap

Controlled experiments systematically varying the cognitive state of a conscious observer while measuring the decoherence rate of a nearby quantum system represent a gap in the existing literature in quantum foundations. Decoherence experiments routinely characterise

T_2 against electromagnetic, thermal, vibrational, and magnetic perturbations; none have included the observer's cognitive state as a controlled independent variable verified by EEG. This protocol aims to address that gap, with improved methodology, more precise state characterization via EEG, and a decoherence-focused measurement strategy.

SECTION III

Theoretical Context

3.1 The Von Neumann-Wigner Interpretation

In his 1932 treatise, von Neumann demonstrated that the boundary between the quantum system and the measuring apparatus can be moved arbitrarily far along the measurement chain without affecting the predictions of the theory. The logical terminus of this chain is the conscious observer. Wigner (1961) made the stronger claim explicit: consciousness is necessary for quantum state reduction. Stapp (2007) developed this view further. The present experiment tests a weaker, empirically tractable version: if consciousness participates in state reduction, then varying the cognitive state of a nearby conscious observer should produce measurable differences in decoherence rates.

3.2 Environmental Decoherence Theory

Environmental decoherence theory predicts that the cognitive state of a nearby conscious observer should have no effect on the decoherence rate of a quantum system, provided environmental coupling remains constant. This is the null prediction the experiment is primarily designed to test.

3.3 Orchestrated Objective Reduction (Orch-OR)

Penrose and Hameroff (2014) proposed that consciousness arises from quantum processes in neuronal microtubules, with gravity triggering collapse at a threshold. Orch-OR makes distinct predictions from VNW, though quantitative predictions for this specific experimental configuration have not been derived. This is a limitation.

3.4 Neuroscientific Basis for Meditation as a Controlled Variable

Deep meditation produces neurologically distinctive states well-documented in the peer-reviewed literature. Lutz et al. (2004) reported elevated gamma-band power and long-range phase synchrony in long-term meditators. Berkovich-Ohana et al. (2012) documented EEG correlates of mindfulness-induced changes in gamma band activity, including DMN deactivation. Winter et al. (2020) provided further characterization of neural markers associated with content-free awareness states.

Key features include: gamma wave coherence, default mode network suppression (Brewer et al., 2011), distinct differentiation from sleep, and structural brain changes. EEG identifies neural correlates consistent with these states, not the subjective states themselves. EEG provides convergent evidence for the target state, not definitive verification of subjective experience.

3.5 T_2 as an Operational Proxy

The VNW interpretation concerns wavefunction collapse in a measurement context, while T_2 measures the loss of off-diagonal coherence in a quantum system's density matrix. These are not the same process, and a null on T_2 does not rule out a VNW effect in pure measurement-outcome statistics. Decoherence time is used here as an operational proxy for

any putative consciousness-coupling channel — not as the canonical VNW observable. The reasoning is that if consciousness modulates state reduction in any way, the quantum-to-classical boundary is the natural place to look for leakage, and T_2 is one experimentally accessible handle on that boundary. The secondary QRNG arm (§VII) provides an independent test in the measurement-outcome channel, so the protocol does not depend on a single operationalisation.

SECTION IV

The Mechanism Problem

No confirmed physical mechanism currently exists by which the cognitive state of a conscious observer would alter the decoherence rate of a nearby quantum system beyond the known effects of environmental decoherence. The physical coupling — if it exists — is not specified by any current theory at a level that yields quantitative predictions for this experimental configuration.

This limitation is stated directly. The VNW interpretation posits a role for consciousness in state reduction but does not specify the mechanism. Orch-OR proposes a specific mechanism but has not derived quantitative predictions for the effect of a meditator on a separate quantum system. No existing theory provides a calculation of the expected effect size, the optimal sensor, the distance dependence, or the specific cognitive state that would maximize any such effect.

4.1 The Artifact-Prior Objection, and Why the Design Defeats It

It is sometimes argued that, in the absence of a mechanism, any positive signal from this kind of experiment should be read as a sensor artifact or shielding failure rather than as evidence of a new coupling channel. That argument applies to a single-condition experiment. It does not apply to this design. The protocol runs seven conditions in randomised order with synchronous recording on independent environmental channels (SQUID magnetometers, fluxgate magnetometer, temperature sensors, accelerometer), with EEG verification of cognitive state throughout. A shielding failure or sensor drift produces variation that tracks time, temperature, mains frequency, or building activity — not the randomised condition schedule.

For an artifact to mimic the predicted result, it would have to vary in lock-step with EEG-verified cognitive states across randomised trials while leaving every monitored environmental channel flat. That is not how shielding failures behave. The randomised condition schedule and the independent environmental monitoring together convert “is this an artifact?” into a question the data can answer directly.

4.2 Scope of the Experiment Given the Mechanism Gap

Because no current theory predicts a quantitative effect size, distance dependence, or optimal cognitive state, the experiment is pre-registered as an exploratory search for statistical signals across a defined set of conditions, with pre-registered thresholds for what will count as a candidate signal, what will count as a null, and what upper-bound effect sizes the protocol rules out if no signal appears. Null results contribute concrete value in the form of pre-registered upper bounds on consciousness-dependent state reduction. Positive results require independent replication on different hardware, in a different facility, before they can be claimed as evidence of a new coupling channel.

SECTION V

Experimental Design

5.1 Multi-State Consciousness Conditions

The multi-state design addresses a key methodological critique of prior consciousness-quantum research: the relationship between specific meditation styles and observer effects is not empirically established. This design allows the data to determine which state — if any — produces the effect, rather than assuming the answer in advance.

The seven conditions were chosen to span the full cognitive-excitation axis and are presented below in descending order of cognitive excitation. No structural significance is placed on the number itself.

Condition	Description	EEG Markers	Rationale
Control 3: Active Cognitive Task	Meditator performs mental arithmetic or verbal tasks	High beta, active frontal regions	Controls for cognitive engagement — the “noisy mind” condition
Control 2: Non-Meditator Resting	Untrained person sits in chamber, eyes closed, relaxed	Normal resting-state EEG (alpha dominant)	Controls for mere human presence, body heat, EM from nervous system
Condition A: Focused Attention (Shamatha)	Experienced meditator in concentrated focus on single object (breath)	Increased beta/gamma coherence, frontal midline theta	Tests whether concentrated attention affects decoherence
Condition B: Open Monitoring (Vipassana)	Experienced meditator in receptive awareness, no specific focus	Increased theta, decreased beta, parietal changes	Tests whether open receptive awareness affects decoherence
Condition D: Non-Directed Openness	Experienced contemplative practitioners in complete receptive surrender without specific technique	High alpha/theta coherence, heart-brain synchronization, sustained low beta power, without the gamma burst or EEG flattening that defines objectless awareness / cessation	Tests whether non-technique-based receptive openness produces different effects than Condition C
Condition C: Objectless Awareness / Cessation	Experienced meditator in contentless awareness or full cessation (nirodha)	Gamma burst followed by EEG flattening, or sustained high-amplitude gamma	Tests the primary hypothesis — maximum cognitive silence
Control 1: Empty Chamber	No human present	N/A	True baseline — eliminates all human variables

Table 1: The seven controlled consciousness conditions, ordered by descending cognitive excitation.

Note on Condition D. The framework suggests the mechanism may involve receptive openness rather than technique-based cognitive reduction. If the effect is mediated through receptive openness rather than through progressive silencing, Condition D might produce

different results than Condition C. The multi-state design allows this question to be answered empirically.

5.1.1 Why the Multi-State Design Works

The seven-condition structure is what makes this experiment interpretable in either direction, and it is what separates the protocol from prior consciousness-quantum work. Five properties of the design follow:

It changes the question being asked. A single-condition test asks whether T_2 is different in a meditator's presence than in an empty chamber — a question dominated by slow drift in the system. The multi-state design asks whether T_2 varies in a specific ordered pattern across seven randomised conditions. This is a pattern-matching test against the cognitive-excitation axis, pre-registered as a Jonckheere-Terpstra ordered test (§8.2), and there is no environmental confound that produces a signal correlated with EEG-verified objectless awareness but not with mental arithmetic.

It internalises its own controls. Control 1 (empty chamber), Control 2 (untrained resting), and Control 3 (active cognition) are not separate validation studies — they are conditions in the same randomised block. They calibrate the instrument noise floor, the human-presence baseline, and the high-cognitive-excitation endpoint within each session. Prior consciousness-RNG work could not distinguish “intention” from “any focused mental state” because it had only one state. This design rules out broad-arousal, broad-cognitive-activation, and mere-human-presence explanations because all three are explicitly conditioned on.

Each condition is a falsifier for the others. Because no current theory specifies which cognitive state should produce the effect — focused attention, open monitoring, cessation, non-directed openness all have claims in the contemplative literature — the protocol does not commit. It tests all four against three controls and lets the shape of the result discriminate between theories. If only Condition C deviates: a VNW-style cessation effect. If A-D scale with gamma coherence: a dose-response. If D differs from C: evidence for receptive openness over progressive silencing. If all conditions are flat: environmental decoherence theory is supported and a pre-registered upper bound is established.

It defeats the artifact-prior objection structurally. As discussed in §4.1, for a shielding failure or sensor drift to mimic the predicted result it would have to vary in lock-step with EEG-verified cognitive states across randomised trials while leaving every synchronously monitored environmental channel flat. Shielding failures do not behave that way; they track time, temperature, mains frequency, building activity. The randomised condition schedule and the independent environmental monitoring together turn the artifact question into one the data can answer directly.

It turns the mechanism gap into a design feature. Because the protocol does not require a quantitative prediction from any specific theory, it is not falsified by the absence of one. The structure means a null result bounds something specific, and a positive result discriminates between specific theories. That is the property the adjacent literature consistently failed to achieve.

5.2 The Shielded Chamber and Five-Layer Isolation

The chamber minimizes all known sources of decoherence so that any residual variation across conditions cannot be attributed to uncontrolled environmental factors. The chamber is a 3.0 m × 3.0 m × 3.0 m cube with a dual-door airlock entry system on the east face. The isolation system consists of five functional layers, with the magnetic shielding implemented as a dual-layer subsystem (Layer 2a inner, Layer 2b outer) separated by a 50 mm air gap that increases effective magnetic shielding by approximately 3× compared to a single shell of equivalent total thickness. The dual-layer mu-metal architecture is standard practice for MEG, SQUID, and precision-magnetometry chambers. See TSP-ENG-001 Rev B for full engineering specifications.

Layer	Technology	Purpose
1 (Inner): Acoustic + Active Thermal	Acoustic foam + aerogel + Peltier-cooled liner (150 mm)	Eliminate airborne noise (>40 dB acoustic); maintain ± 0.01 K via active thermal control
2a: Magnetic — Gradient Suppression	Mu-metal inner shell (3 mm, 80% Ni, hydrogen-annealed)	Combines with Layer 2b for >80 dB DC/LF magnetic attenuation
2b: Magnetic — Primary Attenuation	Mu-metal outer shell (3 mm) + 50 mm air gap to 2a	Dual-layer architecture meets >80 dB target at mains frequency (50/60 Hz)
3: RF / Microwave Shielding	OFHC copper sheet (2 mm), welded seams	Eliminate external EM radiation (>100 dB, 10 kHz – 10 GHz); pre-attenuates RF before it reaches mu-metal
4: Vibration Isolation	Pneumatic isolators + neoprene dampers (200 mm)	Sub-nanometer vibration control ($<10^{-6}$ m/s ²)
5 (Outer): Structural / Faraday	Aluminum shell (5 mm) + copper mesh cage (245 mm total)	Full EM isolation (>120 dB combined) and mechanical rigidity

Table 2: Five-layer isolation specifications (Rev B architecture). Layer 2 is implemented as a dual-shell mu-metal subsystem (2a inner / 2b outer) separated by a 50 mm air gap, both sitting inside the copper RF shell. Total shielding thickness: 650 mm per side.

For Phase 1, deep underground location ($\geq 1,000$ m rock overburden) provides cosmic ray attenuation $\geq 10^6$. Recommended facilities: SURF, SNOLAB, LNGS, or Boulby — though see §11.4 on facility-access constraints.

5.3 Quantum Sensor Options

Parameter	Option A: Superconducting Transmon Qubit	Option B: NV-Center in Diamond
T_2 (decoherence time)	50-200 microseconds	1-10 milliseconds at room temperature
Operating temperature	~15 mK (dilution refrigerator)	Room temperature
Cost (sensor + readout subsystem)	~\$400K-\$800K (cryostat + transmon + microwave electronics)	~\$30K-\$80K (diamond, optics, microwave control, APD)
Sensitivity	Highest sensitivity to environmental perturbation	Lower sensitivity, longer baseline T_2
Advantages	Maximum detection sensitivity	Cheaper, simpler, no cryogenics
Disadvantages	Expensive, complex, requires specialized operators	Less sensitive to small perturbations

Table 3: Quantum sensor options for decoherence measurement. Costs shown are for the sensor and readout subsystem only and are consistent with the TSP-ENG-001 Rev B Bill of Materials; full-system Phase 0 and Phase 1 budgets in §11 include shielding, support systems, and labor.

Recommendation: NV-center in diamond for Phase 0. If a positive signal is detected and replicated, upgrade to superconducting qubit for Phase 1.

5.4 Instrument Array and Environmental Monitoring

Instrument	Measures	Target	Sensitivity
Quantum Coherence Sensor (NV or transmon)	Decoherence rates (T_2)	Primary experiment	$\leq 1 \mu\text{s}$ resolution
QRNGs (×4)	Quantum randomness statistics	Secondary experiment	$\geq 1 \text{ Mbit/s}$, quantum-sourced
64-Channel Wireless EEG	Neural state characterization	Independent variable	$\geq 1 \text{ kHz}$ sampling
SQUID Magnetometers	Magnetic fields (femtotesla)	Environmental monitor	$< 10 \text{ fT}/\sqrt{\text{Hz}}$
Temperature Sensors (×4)	Thermal environment inside shield	Artifact control	$\pm 0.001 \text{ }^\circ\text{C}$ resolution
3-Axis Fluxgate Magnetometer	Ambient magnetic field inside shield	Artifact control	Continuous recording
3-Axis Accelerometer	Vibration inside chamber	Artifact control	10^{-9} g , continuous

Table 4: Instrument array. All environmental channels recorded synchronously with quantum sensor data and EEG via GPS-disciplined 10 MHz common time base ($< 1 \mu\text{s}$ inter-channel synchronization).

If any anomaly appears in quantum data, environmental channels confirm whether it correlates with a physical artifact or is specific to the consciousness condition. This is the most critical step in the artifact analysis.

5.5 Session Protocol (Revised)

All sessions follow a strict single-observer rule.

5.5.1 Garment and contamination control. Observers wear only facility-provided non-metallic garments. No jewelry, conductive materials, synthetic fibers, or external devices are permitted inside the chamber. All personal items are removed and stored in the external preparation area prior to airlock entry.

5.5.2 Airlock contamination check. Upon entering the airlock, the observer remains for a 60-second stabilization period during which an RF power meter confirms residual electromagnetic emission below -80 dBm (10 kHz – 10 GHz) and a fluxgate magnetometer confirms no ferromagnetic contamination. Only after the contamination check passes is the inner door released.

5.5.3 Strict single-observer rule. Only one observer is permitted inside the main cubic volume during any active data-collection session. No assistants, additional personnel, or equipment beyond the pre-installed sensor array are present.

5.5.4 Session timing. Session structure differs between phases:

Phase 0 (single-condition sessions): Each session consists of one consciousness condition only. Condition period is 30 minutes regardless of which condition is run. No inter-condition buffers are required because no condition transitions occur within a session. Each session is preceded by empty-chamber calibration and observer settling and followed by a brief post-session baseline. One session per observer per 24 h.

Phase 1 (chained multi-condition sessions): A single session may run all seven conditions in randomized order. Condition periods are 15 minutes each for Conditions 1, 2, 3, A, B, and D; Condition C (cessation) is 30 minutes. A 10-minute buffer separates condition transitions to allow state reset and minimize carryover effects. Total session duration is approximately 3 hours, preceded by empty-chamber calibration and observer settling and followed by a brief post-session baseline. One session per observer per 24 h.

5.5.5 Emergency procedures. Inner-door panic bar allows immediate egress. Fiber-optic video and intercom monitoring throughout. At no point during normal operation are both airlock doors open simultaneously.

5.6 Participants

- Conditions A, B, C: minimum 5 experienced meditators with documented ability to reliably produce EEG patterns consistent with the target states (gamma coherence + DMN suppression) under laboratory conditions.
- Condition D: minimum 3 experienced contemplative practitioners with documented practice in non-directed receptive openness.
- Control 2: minimum 5 non-meditators matched for age and sex.
- Control 3: same meditators as Conditions A–C (within-subjects comparison).
- Sessions: each participant completes a minimum of 50 sessions per condition, randomized order.

5.7 Blinding and Replication

The protocol is pre-registered, randomised, and outcome-blinded where feasible. Statistical analysis is fully blinded by an independent statistician who does not see condition labels until after the analysis is locked. The subject does not know which instruments are active; instrument operators do not know the condition schedule; initial data analysts identify anomalous periods independently, which are then correlated with session logs by a separate team. Subject blinding is not possible for consciousness-state experiments — a meditator knows when they are meditating. This limitation is inherent to the design and is mitigated by EEG verification of the cognitive state (so the independent variable is operationalised by an objective measurement rather than self-report), by randomisation of condition order, by synchronous environmental monitoring (§5.4), and by the independent blinded analysis described above.

Full replication: the protocol is intended to be repeated with different meditators, different instruments of the same type, and different facilities.

SECTION VI

Primary Experiment: Quantum Decoherence Test

The quantum decoherence test is the primary experiment.

6.1 Logic

If consciousness participates in wavefunction collapse (VNW), then the cognitive state of a nearby conscious observer should affect decoherence rates. The predicted VNW ordering is $T_{\text{coherence}}(\text{cessation}) > T_{\text{coherence}}(\text{baseline}) > T_{\text{coherence}}(\text{active cognition})$. The multi-state design allows testing of which meditation state, if any, produces the largest effect.

6.2 Theory Discrimination

Theory	Predicted Pattern	Key Discriminator
Von Neumann-Wigner	Cessation > Baseline > Cognition	Consciousness state affects decoherence rates
Environmental Decoherence	All conditions equal	No variation across any conditions
Orch-OR	Meditation ≠ Cognition (pattern TBD)	Consciousness affects local spacetime geometry
Epiphenomenalism	All conditions equal	Identical to decoherence prediction

Table 5: Adversarial theory-testing matrix.

SECTION VII

Secondary Experiment: Quantum Random Number Generator Test

A meta-analysis by Bösch, Steinkamp, and Boller (2006) of prior consciousness-random-number-generator research found that reported effect sizes correlated inversely with methodological quality, which is the classic signature of systematic bias. The present protocol does not build on any positive claim from that literature. It uses a QRNG channel only as a secondary, independent test, with underground shielding (in Phase 1), four independent quantum-source generators ($\geq 10^9$ bits per session), pre-registered and randomised condition assignment, outcome-blinding where feasible, fully blinded statistical analysis by an independent statistician, and EEG-characterised cognitive states rather than diffuse “intention.” The QRNG test is independent of the decoherence test. A positive result on one with a null result on the other would carry distinct theoretical implications and is one reason the protocol does not rely on a single operationalisation (see §3.5).

SECTION VIII

Data Analysis and Statistical Framework

8.1 Pre-Registration

Complete analysis plan pre-registered on OSF or AsPredicted before any data collection. Includes: primary hypothesis, secondary hypotheses, planned sample size, statistical tests, stopping rules, correction for multiple comparisons, and criteria for conclusions.

8.2 Primary Analysis (one pre-registered test)

Primary test: the ordered prediction $T_{\text{coherence}}(\text{cessation}) > T_{\text{coherence}}(\text{baseline}) > T_{\text{coherence}}(\text{active cognition})$, tested with a pre-registered ordered test (Jonckheere-Terpstra or equivalent) at $p < 0.001$.

8.3 Secondary Analyses (at most three pre-specified)

- Cessation (Condition C) vs. Control 1 (empty chamber) — paired comparison, Bonferroni-corrected.
- Dose-response within meditation conditions: do sessions with deeper EEG markers (higher gamma coherence, stronger DMN suppression) show larger T_2 effects?
- All-meditation-conditions pooled vs. all-controls pooled.

Bayesian analysis runs in parallel on the primary and secondary tests (Bayes factors; $BF > 10$ = strong evidence, $BF > 100$ = decisive).

8.4 Exploratory Analyses

Any other cross-condition comparison, any subgroup analysis, any post-hoc pattern detection, and all QRNG-only analyses are labeled exploratory. Exploratory results are hypothesis-generating only and are not used to claim a consciousness-dependent effect.

8.5 Artifact Analysis

For every significant result, examine all environmental channels for correlated changes. If T_2 correlates with any environmental variable (temperature, magnetic field, vibration), classify as artifact. A result is only classified as a candidate consciousness effect if no environmental correlate is found. See §4.1 for the general logic of why this test, combined with randomised condition ordering, separates artifact from candidate signal.

SECTION IX

Power Analysis

A conventional power analysis requires an expected effect size derived from theory or prior data. For the decoherence test, neither is available. Phase 0 can detect medium-to-large effects ($d > 0.3$); it cannot detect the very small effects claimed in older RNG literature. A Phase 0 null result does NOT mean the effect does not exist — it means it is smaller than Phase 0 can detect. Phase 1, with increased trial counts and more sensitive instrumentation, would be required to test for smaller effects.

SECTION X

Stopping Rules

Criterion	Action
Phase 0: 500 total sessions minimum completed	Proceed to analysis
All $p > 0.10$	Stop. Publish null result with pre-registered upper bounds.
Marginal signal ($0.001 < p < 0.05$)	Extend to 1,500 sessions for confirmation
Strong signal ($p < 0.001$ with clean environmental channels)	Proceed to Phase 1 (subject to host-collaboration adoption — see §11.2)

Table 6: Pre-registered stopping rules.

Data collection continues for the pre-specified number of sessions regardless of interim results. No optional stopping.

SECTION XI

Budget, Phasing, and Timeline

11.1 Phase 0 — Foundational Test

Parameter	Specification
Budget	~\$420,000–\$470,000 (Full Phase 0 configuration; see TSP-ENG-001 Rev B Bill of Materials)
Location	University lab or dedicated experimental facility with purpose-built shielded chamber
Quantum sensor	NV-center in diamond
EEG	64-channel system, optically isolated, battery-powered with fiber-optic data egress
QRNG	4× quantum-source units, fiber-coupled
Shielding	Full dual-mu-metal + copper + Faraday architecture per TSP-ENG-001 Rev B
Personnel	PI + 1 postdoc + meditator participants (voluntary)
Timeline	12 months
Parameter	Specification
Primary scientific goal	Pre-registered seven-condition test of the von Neumann-Wigner interpretation against environmental decoherence theory, using EEG-verified cognitive state as the independent variable and T_2 on an NV-center as the dependent variable. Go / no-go for Phase 1.
Co-product	Sensor-and-shielding characterisation paper documenting residual environmental coupling to T_2 under the chamber's full isolation regime. Independently publishable in a metrology venue regardless of consciousness-arm outcome.

Table 7: Phase 0 specifications.

The Phase 0 figure reflects the full engineering scope of TSP-ENG-001 Rev B (dual mu-metal, active thermal control, LTS SQUID environmental monitoring, optically-isolated EEG). Two reduced configurations are available if funding requires staging: a Minimal Phase 0 (~\$140K–\$210K) using a single mu-metal layer, 1.5 m cube, NV + QRNG only and no SQUID; and a Micro-Pilot (~\$2K–\$6K) using a commercial QRNG only with no shielded chamber. The Micro-Pilot is also described separately in Appendix A as personal exploratory activity outside the pre-registered protocol. The figures above are stated as USD equivalents of the TSP-ENG-001 Rev B Euro-denominated Bill of Materials at prevailing exchange rates and are intended as planning estimates; final figures will track procurement quotes at the time of funding.

NV-centers are characterised in lab cryostats every day; the engineering required for this protocol is not. The chamber's specifications are dictated entirely by what is required to make the consciousness arm interpretable. The metrology paper is what falls out of doing the consciousness test correctly, not the other way around.

11.2 Phase 1 — Conditional Follow-up

Parameter	Specification
Budget	~\$1M–\$3M (full system including dilution refrigerator, transmon platform, underground deployment, and 2–3 years of operation)
Location	Underground laboratory (SURF, SNOLAB, LNGS, or Boulby)
Quantum sensor	Superconducting transmon qubit system
Shielding	Full five-layer isolation with airlock
Participant pool	Larger pool, extended data collection
Timeline	2–3 years
Adoption requirement	Phase 1 facility access is allocated through established collaborations and is not realistically available to an unaffiliated researcher. The only viable path to Phase 1 is via a host collaboration that adopts the protocol after Phase 0 results justify scaling up.
Goal	Definitive result with adequate statistical power

Table 8: Phase 1 specifications. The Phase 1 figure reflects realistic full-system cost: the dilution refrigerator alone is approximately \$235K–\$585K (TSP-ENG-001 Rev B §8.2), and the full transmon platform, underground deployment costs, and multi-year operation drive the total into the \$1M–\$3M range. Phase 1 is conditional on Phase 0 results and on adoption by an institutional host.

11.3 Cost Comparison

Experiment	Cost
Large Hadron Collider (CERN)	\$13.25B
James Webb Space Telescope	\$10.0B
LIGO	\$1.1B
LUX-ZEPLIN (LZ)	\$55M
Silence Experiment Phase 1	\$1M–\$3M
Silence Experiment Phase 0	\$420K–\$470K

Table 9: Cost comparison with major physics experiments.

11.4 Recommended Facilities (Phase 1)

Facility	Location	Depth	Notes
SURF (Primary)	Lead, South Dakota	1,490 m (~4,300 m.w.e.)	Hosts LZ; US-based
SNOLAB	Sudbury, Ontario	2,070 m (~6,010 m.w.e.)	Deepest in N. America
LNGS	Gran Sasso, Italy	1,400 m (~3,800 m.w.e.)	Largest underground lab
Boulby	North Yorkshire, UK	1,100 m (~2,850 m.w.e.)	Hosts ultra-low-background physics experiments

Table 10: Recommended underground physics facilities (Phase 1).

Slots at these facilities are allocated through established collaborations. An unaffiliated researcher cannot access them directly; Phase 1 proceeds only via a host collaboration that adopts the protocol following Phase 0 (see §11.2).

11.5 Timeline

Phase 0: Months 1–3 facility negotiation and equipment procurement; Months 4–6 chamber construction, calibration, meditator screening; Months 7–11 data collection across conditions; Month 12 blinded analysis and manuscript preparation. **Phase 1** (if warranted by Phase 0 results and a host collaboration is in place): approximately 2–3 years from Phase 0 completion.

11.6 Peer-Review Submission Plan

This protocol is intended for submission as a Registered Report to a journal that accepts the format (e.g. *Royal Society Open Science*, *PLOS ONE*, *Collabra: Psychology*), with the protocol reviewed before data collection begins. Both this paper and the companion theoretical paper will also be posted to arXiv (quant-ph or physics.gen-ph). Zenodo and OSF provide stable archiving and pre-registration but are not a substitute for peer review; the current papers are labelled as concept papers not yet peer-reviewed.

SECTION XII

Ethical Considerations

IRB approval is required before data collection. The protocol monitors for claustrophobia, anxiety, and adverse meditation experiences (e.g., dissociation, distress). Informed consent includes disclosure of the speculative nature of the hypothesis, the possibility of null results, and the investigator's theoretical commitment to the framework. Data anonymisation procedures are specified. All data and analysis code will be deposited in an open repository upon publication. The investigator's theoretical commitment to the Silence Paradigm framework is disclosed as a potential source of bias, mitigated by the pre-registered, randomised, outcome-blinded protocol, by fully blinded statistical analysis by an independent statistician (see §5.7), and by the pre-registered statistical plan.

SECTION XIII

Limitations

The following limitations are stated explicitly:

- Phase 0 is underpowered for very small effects ($d < 0.3$). It is a first-stage foundational test, not a definitive result.
- No established physical mechanism exists for consciousness-decoherence interaction. (See §4.1 for why this does not, by itself, force an artifact-only interpretation of any positive signal.)
- T_2 is used as an operational proxy for any putative consciousness-coupling channel; a null on T_2 does not rule out an effect in pure measurement-outcome statistics (see §3.5).
- The multi-state design addresses but does not fully resolve the question of which cognitive state is optimal.
- Underground location (Phase 1) introduces practical constraints on participant access and session scheduling, and Phase 1 is gated on institutional adoption (see §11.2).
- EEG-state correspondence is imperfect — neural markers are correlates, not confirmations of subjective state.
- Subject blinding is not possible for consciousness-state experiments.
- A positive result would establish correlation, not causation or mechanism.
- No quantitative effect-size prediction can be derived from current theory. The experiment is exploratory in its effect-size expectations.

SECTION XIV

Future Directions

The following extensions are speculative and entirely contingent on positive results from the primary experiment: gravitational measurements (atom interferometry) during meditation states; vacuum energy density measurements (Casimir effect plates); precision time-correlation measurements; entangled photon pair monitoring; quantum noise analysis.

Future extensions may explore synchronised multi-observer configurations after the single-observer effect has been robustly established and replicated.

None of these extensions are proposed as part of this experiment.

SECTION XV

Conclusion

This protocol is a foundational test of the von Neumann-Wigner interpretation under controlled observer states — a question that has been part of the quantum literature for nearly a century without ever being tested this way. The primary experiment measures decoherence rates across seven controlled consciousness conditions, testing VNW against environmental decoherence theory and Orch-OR, with one pre-registered primary test and up to three pre-specified secondary tests. All other analyses are exploratory. A sensor-and-shielding characterisation paper is produced as an independently publishable co-product of the same hardware run.

The limitations have been stated directly: no confirmed mechanism, no quantitative effect-size prediction, Phase 0 underpowered for very small effects, Phase 1 conditional on adoption by an institutional host. The protocol is publishable regardless of outcome. A null result establishes pre-registered upper bounds on consciousness-dependent decoherence. A positive result opens a new domain of investigation requiring independent replication on different hardware, in a different facility, before claims can be considered established.

The protocol is intended for submission as a Registered Report (see §11.6). The theoretical basis for this protocol is detailed in the companion paper (Paper 1: The Silence Paradigm).

APPENDIX A

Personal Exploratory Activity (Not Part of the Main Protocol)

A small-scale commercial-QRNG exploration (~\$2,000-\$5,000) is conducted as personal exploratory activity outside the main protocol. It is described here for transparency only.

Parameter	Specification
Budget	~\$2,000-\$5,000
Equipment	Commercial QRNG only (no quantum coherence measurement)
Location	Any quiet room with basic shielding
Participants	1-2 experienced meditators
Sessions	200+ per condition
Goal	Personal exploration of methodology and feasibility. NOT a test of the primary hypothesis and not a pre-registered component of Phase 0 or Phase 1.

This activity is explicitly outside the formal protocol. Its results are not pre-registered, are not used to claim effects, and are not combined with Phase 0 or Phase 1 data in any confirmatory analysis.

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